Which database should we use for product analytics: PostgreSQL, ClickHouse or BigQuery?

Choose PostgreSQL with the TimescaleDB extension. It covers 90% of the analytics workload without a second database and keeps operations simple for a team of three.

Neden PostgreSQL? (Türkçe özet)

Ekip küçük; tek bir veritabanı işletmek, iki ayrı sistem işletmekten çok daha ucuzdur. Günlük 40 milyon satıra kadar TimescaleDB yeterli performans gösterir.

Capacity estimate

Expected load is $\mathrm{N} = 4 \times 10^{7}$ rows per day; with compression ratio $\mathrm{k} \approx 8$ storage for a year is

$$\mathrm{S} = (\mathrm{N} \cdot 365 \cdot 120 \text{ bytes}) / \mathrm{k} \approx 220 \text{ GB}$$

Benchmark summary

Engine	Ingest, rows/s	p95 query, ms	Storage/yr	Ops effort	License	Managed in RU	SQL dialect	Joins
PostgreSQL + TimescaleDB	180 000	420	220 GB	low	Apache/TSL	yes	full SQL	full
ClickHouse	1 200 000	85	90 GB	medium	Apache 2.0	yes	ClickHouse SQL	limited
BigQuery	n/a (batch)	1 900	pay per TB	very low	proprietary	no	GoogleSQL	full

Migration snippet

```
CREATE EXTENSION IF NOT EXISTS timescaledb;
SELECT create_hypertable('events', 'created_at', chunk_time_interval ⇒ INTERVAL '1 day');
ALTER TABLE events SET (timescaledb.compress, timescaledb.compress_segmentby = 'tenant_id');
```

Documentation: https://docs.timescale.com/use-timescale/latest/compression/about-compression/?utm_source=council-report&utm_medium=pdf&utm_campaign=very-long-link-check

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